

19. To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events.

To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling.

```
#include <stdio.h>
#include <stdlib.h>
#include <errno.h>
#include <string.h>

int main() {
    FILE *fp;

    fp = fopen("missing.txt", "r");

    if (fp == NULL) {
        fprintf(stderr, "Runtime Error: %s\n",
                strerror(errno));

        FILE *log = fopen("error.log", "a");

        if (log != NULL) {
            fprintf(log,
                    "Error: Unable to open missing.txt\n");
            fclose(log);
        }

        printf("Program recovered gracefully.\n");

        return 0;
    }

    fclose(fp);

    return 0;
}
```

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano skill21.c  
bindu@Bindusree:~$ gcc skill21.c -o skill21  
bindu@Bindusree:~$ ./skill21  
Runtime Error: No such file or directory  
Program recovered gracefully.
```